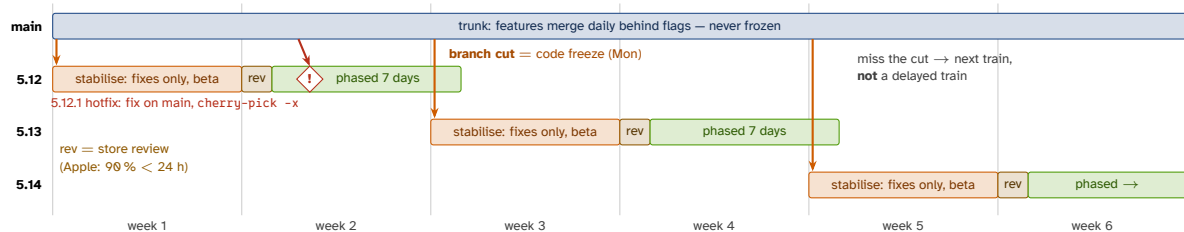


In one line: A mobile release cannot be undone: every build is **reviewed, signed**, rolled out **gradually**, and then lives on phones for months. So releases leave on a fixed **train** (date fixed, scope floats), each build gets a **strictly increasing build number**, the backend stays **compatible with every version still installed**, a **minimum-version switch** exists before it is needed, and “rollback” means **roll forward**: a flag flip or a hotfix build.

A two-week release train



Version vs build number

	iOS	Android
shown	CFBundleShortVersionString: three integers, 5.12.1	versionName: any string; the only value users see
build	CFBundleVersion: 1-3 integers; unique per upload of a version	versionCode: positive int, never reused, max 2100 000 000
order	build must rise for each upload of the same version	install refuses a <i>lower</i> versionCode: no downgrades

Set the build number from CI (run number or timestamp), never by hand; tag every build (v5.12.1+4211) so a crash's build maps to a commit.

How a train runs

- **Fixed cadence** (1–2 weeks): the date is fixed, the scope floats; unfinished work ships *dark* behind a flag. Small batches = small risk, predictable dates for product and QA.
- **Branch cut / code freeze**: release/5.12 from main; only approved fixes land, each fixed on main first and cherry-picked. Regression suite, beta (TestFlight / Play testing track), release notes.
- **Go / no-go** by a rotating **release captain**: crash-free and hang rates of the beta, open blockers, backend ready, store assets done. Then submit, release manually after approval.
- **Rollout**: App Store phased release (automatic updates only, 7 days) · Play staged rollout (you pick the %, can **halt**). Gate each step on crash-free users vs the previous version.

Old versions never die — backward compatibility

- Every version since the **minimum supported** one is a client of your API: users with auto-update off, devices stuck on an old OS. Send app version + build in a header and *measure* them.
- API changes are **additive** (expand → migrate → contract): add a field, never rename or change its meaning; clients ignore unknown fields and map unknown enum cases to .unknown. Breaking change = new endpoint / API version, old one kept until its traffic is negligible.
- **Min supported version**: remote config returns minVersion (below → **forced** update, blocking screen → store) and recommendedVersion (below → **soft** prompt). Android also has Play **In-App Updates** (immediate / flexible); iOS has no system API — compare yourself.

Remember

“Trains leave on time, builds only go up, old apps never die — ship the min-version switch in v1.”

Example — forced vs soft update

```
struct UpdatePolicy: Decodable { // from remote config
    let minVersion: String; let recommendedVersion: String }
enum UpdateState { case ok, soft, forced }
func parts(_ s: String) -> [Int] { // "5.9" -> [5, 9, 0]
    Array((s.split(separator: ".").map { Int($0) ?? 0 }
        + [0, 0, 0]).prefix(3)) }
func state(current: String, _ p: UpdatePolicy) -> UpdateState {
    let c = parts(current) // numeric, not "5.9" > "5.12"
    if c.lexicographicallyPrecedes(parts(p.minVersion)) { return .forced }
    if c.lexicographicallyPrecedes(parts(p.recommendedVersion)) { return .soft }
    return .ok }
// current: Bundle.main.infoDictionary?["CFBundleShortVersionString"]
```

There is no rollback

Neither store lets you push an older build: Android refuses a lower versionCode, the App Store serves only the latest approved version. Halting a phased/staged rollout only protects users *not yet updated*. The levers, fastest first: **server-side kill switch / flag** → backend fix → **hotfix build** (or the old code rebuilt with a *higher* build number). RN: an OTA JS update if runtime-compatible.

Hotfix process

Triage severity (crash rate, data loss, security, revenue) → mitigate server-side → fix on main with a test → cherry-pick to release/5.12 → 5.12.1, new build → Apple **expedited review** for a critical bug fix → faster rollout → postmortem.

Keys and notes

- **Signing custody**: iOS distribution identity in an encrypted store (match) + CI secrets, few humans. Android **Play App Signing**: Google holds the *app signing key*, you hold the *upload key* (a lost one can be reset); new apps are enrolled automatically.
- **Release notes**: user-facing “What’s New” per locale, written for users; the internal changelog comes from PR labels / conventional commits.

Traps

- ! Adding the forced-update check *after* you need it — the old binary doesn't have it.
- ! Comparing versions as strings: "5.9" > "5.12".
- ! Renaming a JSON field used by a version still on 8 % of devices.
- ! “Phased release lets us roll back” — it pauses, it does not undo.